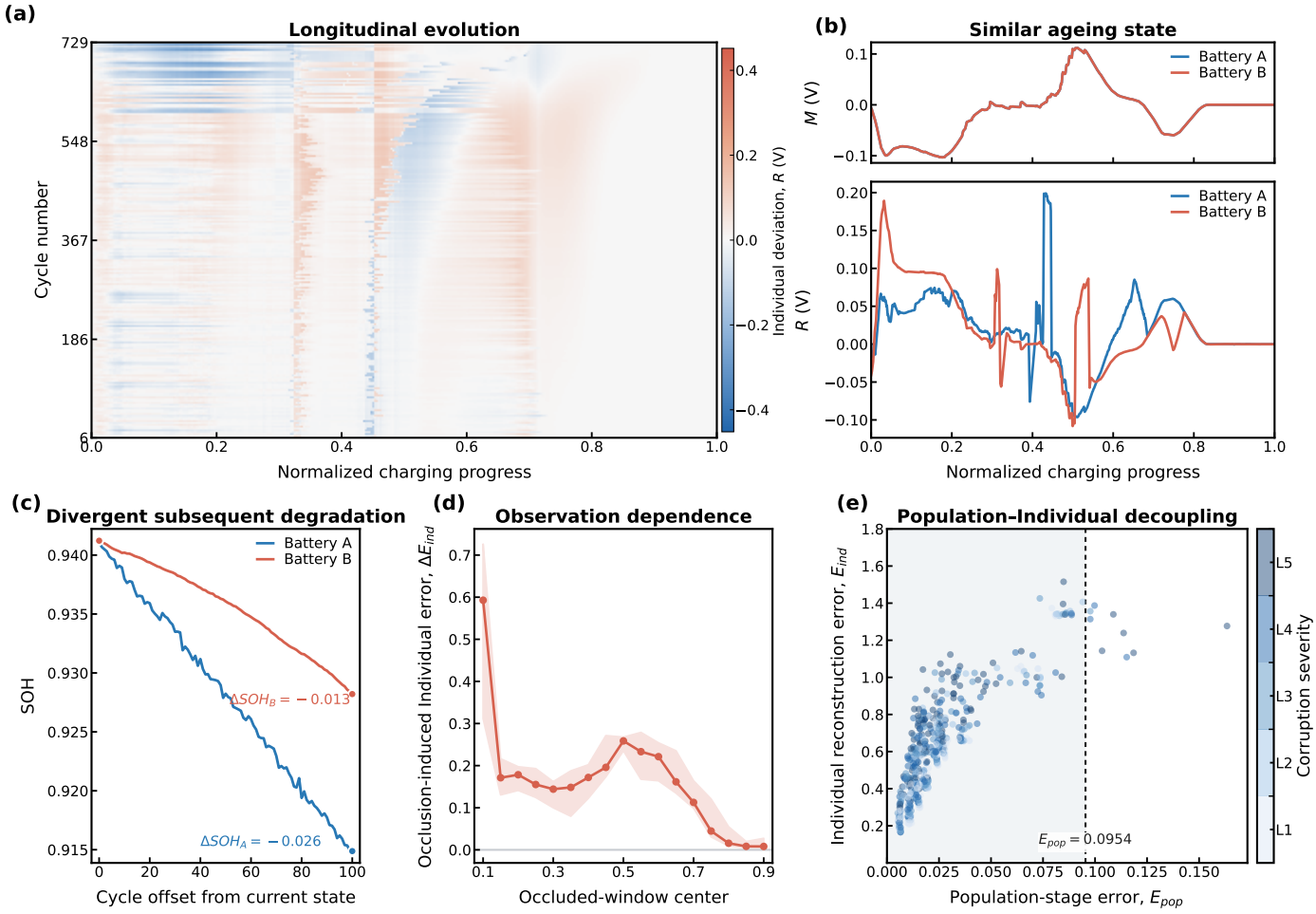

(a)
Longitudinal evolution
Cycle number
729
548
367
186
6
0.0
0.2
0.4
0.6
0.8
1.0
Normalized charging progress
Individual deviation, R (V)
0.4
0.2
0.0
−0.2
−0.4
(b)
Similar ageing state
M (V)
0.1
0.0
−0.1
Battery A
Battery B
R (V)
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
Normalized charging progress
(c)
Divergent subsequent degradation
SOH
0.940
0.935
0.930
0.925
0.920
0.915
Battery A
Battery B
ΔSOH_B = −0.013
ΔSOH_A = −0.026
0
20
40
60
80
100
Cycle offset from current state
(d)
Observation dependence
Occlusion-induced Individual error, ΔE_ind
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.1
0.3
0.5
0.7
0.9
Occluded-window center
(e)
Population–Individual decoupling
Individual reconstruction error, E_ind
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
E_pop = 0.0954
0.000
0.025
0.050
0.075
0.100
0.125
0.150
Population-stage error, E_pop
Corruption severity
L5
L4
L3
L2
L1